專題報告 2

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演講標題： Presentation title

Recent Heat-Related Illness Prevention Measures in Japan

近期日本熱危害引起相關疾病預防方法

Abstract

In 2025, the number of workers in Japan who experienced occupational heat-related illness resulting in four or more days of absence reached a record high of 1,803, including 19 fatalities.

Occupational heat-related illness prevention in Japan focused primarily on high-temperature workplaces where heat sources were present. However, following a severe heatwave in 1994 that resulted in more than 20 occupational fatalities, the government issued guidelines in 1996 recommending preventive measures for a broader range of workplaces, including outdoor work, regardless of the specific type of work performed.

These guidelines have been revised and expanded. In March 2026, the Ministry of Health, Labour and Welfare introduced updated comprehensive guidelines for preventing occupational heat-related illness. In addition, the annual "STOP! Heatstroke Cool Work Campaign," launched annually from 2017, promotes the implementation of preventive measures before and during the summer season. These official initiatives recommend assessing heat exposure by measuring the Wet Bulb Globe Temperature (WBGT) at worksites and implementing appropriate measures, including work environment management, work management, health management, and occupational health education.

Delayed recognition of early symptoms and inadequate initial responses have been identified as major contributing factors to severe and fatal cases. To address these issues, the amended Ordinance on Industrial Safety and Health came into effect in June 2025, requiring employers to implement measures to prevent occupational heat-related illness from progressing to severe illness. Specifically, for work expected to continue for at least one hour or exceed four hours per day in environments where the WBGT is 28°C or higher or the air temperature is 31°C or higher, employers are required to establish reporting systems, develop emergency response procedures, and ensure that relevant personnel are informed of these procedures.

In addition to these regulatory measures, various preventive interventions are being developed to reduce the occurrence of occupational heat-related illness. These include wearable devices designed to identify workers at increased risk, salt-containing candies, pre-cooling strategies, fan-equipped workwear, and ice slurry ingestion. Many of these products have already become widespread in workplaces and are used by numerous workers; however, I believe that while some have been scientifically proven effective for preventing heatstroke, others have not. I report the results of experimental verification regarding the effectiveness of these products.

摘要： (中文翻譯-若與原文語意有出入，以原文為準)

2025 年，日本因職業性熱相關疾病導致四天或以上缺勤的勞工人數創下歷史新高，達 1,803 人，其中包括 19 人死亡。

日本的職業性熱相關疾病預防主要聚焦於高溫工作場所，這些場所存在熱源。然而，1994 年一場嚴重熱浪導致超過 20 人職業死亡後， 政府於 1996 年發布指引，建議針對更廣泛的工作場所採取預防措施，包括戶外工作，無論從事何種特定工作類型。

這些指引已經過修訂與擴充。2026 年 3 月，厚生勞動與福利部推出了更新的綜合性職業性熱相關疾病預防指引。此外，自 2017 年起每年舉辦的「停止！中暑降溫工作運動」(STOP! Heatstroke Cool Work Campaign)，推廣在夏季前及夏季期間實施預防措施。這些官方倡議建議透過測量工作場所的綜合溫度熱指數（WBGT）來評估熱暴露，並實施適當的措施，包括工作環境管理、工作管理、健康管理及職業健康教育。

早期症狀的識別延遲及初步應變處理不足，已被認定為嚴重及致命病例的主要促成因素。為解決這些問題，修訂後的《工業安全與健康條例》(Ordinance on Industrial Safety and Health) 於 2025 年 6 月生效，要求雇主採取措施防止職業性熱相關疾病惡化為重症。具體而言，對於預期每日工作至少持續一小時或超過四小時，且在 WBGT 達到 28°C 或以上或空氣溫度 31°C 或以上的環境下，雇主必須建立通報系統、制定緊急應變程序，並確保相關人員知悉相關程序。

除了這些法規措施外，還有各種預防措施正在開發中，以減少職業性熱相關疾病的發生。這些包括設計用來辨識高風險工人的穿戴式裝置、含鹽糖果、預冷策略、風扇工作服以及冰品攝取。這些產品中許多已經在職場中廣泛使用，並被許多勞工使用；然而，我認為雖然有些產品已被科學證明能預防中暑，但有些則沒有。我報告了這些產品對其有效性的實驗驗證結果。